

# Peilin Cai

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## EDUCATION

### University of Southern Californ

*Master of Science in Computer Science*

Los Angeles, CA

Aug. 2024 – May 2026

### Wuhan University

*Bachelor of Engineering in Computer Science*

Wuhan, Hubei

Sept. 2020 – Jun. 2024

## PUBLICATIONS

### Secure On-Device Video OOD Detection Without Backpropagation

Oct. 2025

*International Conference on Computer Vision (ICCV) 2025 [PDF] — [arXiv:2503.06166]*

Shawn Li; **Peilin Cai**; Yuxiao Zhou; Zhiyu Ni; Renjie Liang; You Qin; Yi Nian; Zhengzhong Tu; Xiyang Hu; Yue Zhao

### A Personalized Conversational Benchmark: Towards Simulating Personalized Conversations

Oct., 2025

*NeurIPS Workshop MTI-LLM Spotlight (Top 5%) 2025 [PDF] — [arXiv:2505.14106]*

Shawn Li; **Peilin Cai**; Ryan A. Rossi; Franck Dernoncourt; ...; Philip S. Yu; Xiyang Hu; Yue Zhao

## RESEARCH EXPERIENCE

### The Earth Simulator: Street View World Modeling with 3D Gaussian Memory and Camera Control

Jun. 2025 – Now

*Research Assistant · USC Viterbi & GVL Lab · Advisor: Prof. Yue Wang*  
*Submitted*

Los Angeles, CA

- Developed a street-view world model that turns a handful of raw, pose-free images into long-horizon, camera-controllable exploration videos grounded in 3D scene geometry.
- Introduced a 3D Gaussian spatial memory decoupled from a video generator, so geometry provides stable structure while a generative prior delivers photorealistic, temporally coherent rollouts.
- Designed a pose-free, self-supervised training pipeline that learns entirely from internet street-view and driving videos, enabling scalability across diverse real-world environments.
- Achieved higher visual fidelity and stronger multi-view and temporal consistency than reconstruction-only, video-only, and existing world-model baselines, supporting interactive virtual exploration and large-scale synthetic data generation.

### LAM: Language Articulated Object Modelers

Feb. 2025 – Sept. 2025

*Co-author · USC Viterbi & iLab · Advisor: Prof. Laurent Itti*  
*Submitted*

Los Angeles, CA

- Co-created a pipeline which formulated articulated-object synthesis as a unified code generation task where geometry and articulations are co-designed from scratch.
- Built an agentic pipeline coordinating *Link Designer* (hierarchy reasoning), *Geometry & Articulation Coders* (code writing, compilation, debugging), and *Geometry & Articulation Checkers* (self-correction).
- Used code as an bridge to select joint types and compute precise placements, ensuring correct inter-link relationships and correct reconstruction of semantics.
- Automatically generated complex, procedurally defined articulated objects from text prompts, demonstrating reliability and scalability.

### A Personalized Conversational Benchmark: Towards Simulating Personalized Conversations (PersonaConvBench)

Feb. 2025 – May. 2025

*Research Assistant · USC Viterbi & FORTIS Lab · Advisor: Prof. Yue Zhao*  
*Submitted*

Los Angeles, CA

- Co-created **PersonaConvBench**, a large-scale benchmark for personalized reasoning and generation in multi-turn conversations. Specifically, the entire technical implementation, data set acquisition and organization, design and execution of all experiments, and multiple iterations of the method.
- Defined three core tasks—sentence classification, impact regression, and user-centric generation—across 10 diverse Reddit-based domains.
- Built unified prompting/evaluation; observed a +198% relative gain in sentiment classification when using personalized history vs. the best non-conversational baseline.
- Released benchmark, evaluations, and code to spur research on long-horizon, user-adaptive LLMs. arXiv:2505.14106

## Secure On-Device Video OOD Detection Without Backpropagation (SecDOOD)

Nov. 2024 – March. 2025

*Research Assistant · USC Viterbi & FORTIS Lab · Advisor: Prof. Yue Zhao*  
*ICCV 2025 (Poster)*

*Los Angeles, CA*

- Co-developed **SecDOOD**, a secure cloud–device collaboration framework enabling on-device OOD detection without device-side backpropagation. Specifically, complete technical implementation, model construction and implementation, and the long period iteration of the model designing.
- Implemented a HyperNetwork-based personalized parameter generator to adapt centrally trained models to device-specific distributions without local fine-tuning.
- Designed dynamic feature sampling & selective encryption to encrypt only the most informative channels, reducing overhead while preserving detection quality.
- Ran extensive experiments across datasets/OOD scenarios; achieved performance comparable to fully fine-tuned baselines. arXiv:2503.06166

## PlayTR: Geometric Layout-Driven Structured Plane Reconstruction of Indoor Scenes

Jun. 2023 – Jun. 2024

*Research Assistant · Wuhan University · Advisor: Prof. Gui-Song Xia*

*Wuhan, Hubei*

- Established a prototype with reference to established planar reconstruction methods. Improve the model reconstruction accuracy and segmentation accuracy in the planar reconstruction task by introducing the encoded 3D layout information and Manhattan world cues from the 2D raw image into the cross-attention Transformer.
- Improved the depth prediction method for planar surfaces in the original method from direct top-down global prediction to weighted sum of depth maps with masks after plane-by-plane prediction.
- Designed the data generation module to extract the conforming 3D Layout trilinear segment sets from the dataset accurately and screen out invalid data.
- Reorganized the evaluation structure, introduce a new depth supervision approach, replace the old methods with the more efficient loss function, and design the Hungarian matching for the new trilinear segment group corresponding to the plane group.
- The segmentation results have been improved by 0.7 percentage points over the original model without replacing the backbone, and the reconstruction results show that we have solved the significant problem of poor contact between planes.

## TECHNICAL SKILLS

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**Languages:** Python, C/C++, JavaScript, HTML/CSS

**Developer Tools:** Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm